Lab Sheet 2 (CST-108)

TCP and UDP Client-server

1) Modify the code of the TCP client and the TCP Echo server given in chapter 5 (Fig. 5.2 to 5.5) of Steven's book to do the following:

(a) TCP client reads the name of a file as command line argument and sends the name to the server.
(b) TCP server opens the file and sends the contents of file to the client
(c) TCP client stores the file (use a different location for storing file if both client and server are running on same machine).
(d) Use diff utility available on Linux to check that two files are same.

Run multiple clients with same file name and show that the files stored by all clients are same.

2) Modify the code of the UDP client and the UDP Echo server given in chapter 8 of Steven's book to do the following:

(a) UDP client reads the name of a file as command line argument and sends the name to the server.
(b) UDP server opens the file and sends the contents of file to the client
(c) UDP client stores the file (use a different location if both client and server are running on same machine).

Run multiple clients with same file name and show that they all store files successfully.

3) Write a program link to implement communication channel. Modify client and server programs written in Q1 above so that UDP client and server communicate through link. Assume that the communication channel loses one out of every 5 packets randomly. Add acknowledgements and timers in UDP client and server to handle lost packets.